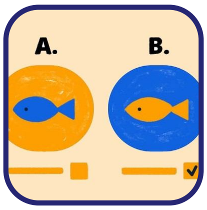
Main approaches

Experimental Methods



Observational Methods

Query Techniques

Physiological Monitoring

1

Evaluation setting

- **laboratory studies**
- **field studies**

2

Laboratory studies

Advantages:
- specialist equipment available (mirrors, computers..)
- uninterrupted environment

Disadvantages:
- lack of context
- difficult to observe several users cooperating

Appropriate
- if system location is **dangerous** or impractical for constrained single user systems to allow controlled manipulation of use
- if we want to **manipulate the context**

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Field studies

Advantages:
- natural environment
- context retained (though observation may alter it)
- longitudinal studies possible

Disadvantages:
- distractions
- noise

Appropriate
- where context is crucial
- for longitudinal studies

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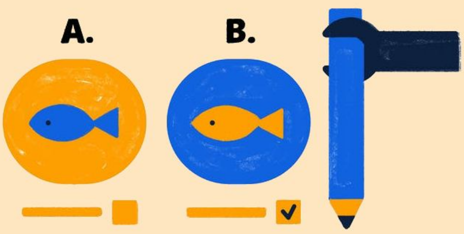
Experimental evaluation

Controlled evaluation of **specific aspects** of interactive behavior.

Controlled experiments provide empirical evidence to support a particular hypothesis.



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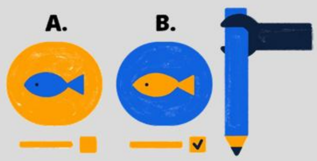
Basic form of an experiment

1. evaluator chooses a **hypothesis** to be tested
2. a number of **experimental conditions** are considered which differ only in the value of some controlled variables
3. changes in behavioral **measures** are attributed to different conditions



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Experimental Factors

Factors that are important to
the overall **reliability** of the experiment:

- **participants** (representative, sufficient sample)
- **variables** (things to modify and measure)
- **hypothesis** (what you'd like to show)

Analyzing results requires a
proper **statistical approach**

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Participants

Participants should **represent** the user population (similar age and level of education)

The **sample size** must be large enough to be considered to be representative of the population, taking into account the design of the experiment and the analysis methods chosen (at least **10 for statistical** analysis)

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Variables

Experiments **manipulate** and **measure** variables under **controlled conditions**.
Two types of variables:

- **independent variables**:
  characteristics **changed** to produce different conditions (e.g. interface style, number of menu items, font sizes)
- **dependent variables**:
  characteristics **measured** in the experiment, influenced only by independent variables
  (e.g. time taken, number of errors)

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Levels (of independent variables)

Each value used for an independent variable is known as level of the variable

If we consider **two independent variables**,
one with **3 levels** and the other with **2 levels**,
we have to consider **3x2 experimental conditions**…

❺❼❾ items ▼▲ command names

❺▼ ❼▼ ❾▼ ❺▲ ❼▲ ❾▲

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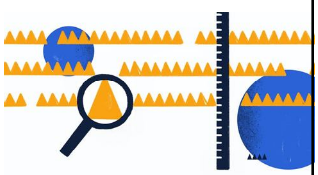
Values of dependent variables

Some **dependent** variables are easy to measure objectively (e.g. time)

Subjective measures can be applied against predetermined **scales**



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Hypotheses

A hypothesis is a prediction of the outcome of an experiment.
It is framed **in terms of the independent and dependent variables** (e.g.. "error rate increases when font sizes decrease")

The **null hypothesis** states that there is no difference in the dependent variable between the levels of the independent variable (e.g. "error rate doesn't change when font sizes decrease")
The aim of the experiment is than to **disprove the null hypothesis**

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About Null Hypothesis

Example

Experimental Hypothesis $\mathbf{H_1}$	**Null Hypothesis** $\mathbf{H_0}$
Smaller fonts increase the error rate	Font size has no effect on the error rate
↓	↓
Goal: prove $\mathbf{H_1}$	Goal: collect enough evidence to reject $\mathbf{H_0}$
The results prove the hypothesis…	*The results support the hypothesis...*

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Next step: decide on the experimental method

Experimental method: **how to assign participants to different conditions**
(which conditions and in which order)

There are **two main methods**:
- **between-subjects** (randomized)
- **within-subjects** (repeated measures)

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Between-subjects method

- each subject performs under **only one condition** (e.g. A or B)
- **no transfer** of learning
- **more users** required
- variation can **bias** results (many testers: better)

A A A B B B

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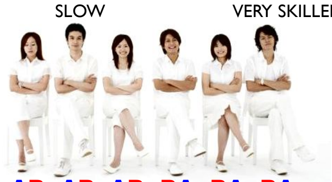
Within-subjects method

- **each subject** performs experiment under more conditions (but **the order is varied** across users)
- **transfer** of learning possible (many testers: better)
- **less costly** (fewer users)
- **less** likely to **suffer from user variation**

SLOW VERY SKILLED



AB AB AB BA BA BA COUNTERBALANCING

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Mixed Designs

When an experiment involves **multiple independent** variables, a popular compromise is a **mixed design**.
In a mixed design:
one variable is tested **between subjects**; ▼▲
another variable is tested **within subjects**. ❺❼❾

❺❼❾ items
▼▲ command names

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Data analysis

Once we have determined
- the **hypothesis** we are trying to test
- the **variables** we are studying
- the **participants** at our disposal
- the **design** that is most appropriate

...we have to decide **how we are going to analyze the recorded results**

Many statistical tests available: if an inappropriate test is chosen, the results can be invalid.

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Few notes on data analysis

Choice of **statistical technique** depends on:

- type of data
- information required (the question we want to answer)

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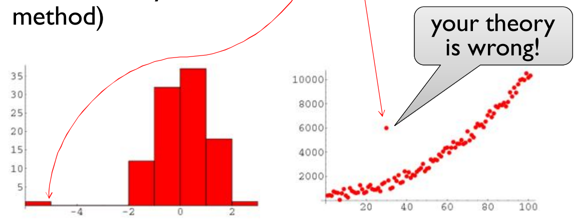
Analysis of data

Before you start to do any statistics:

- look at data (graphs.., **outliers**..)
- **save original data** (as we may later want to try a different analysis method)



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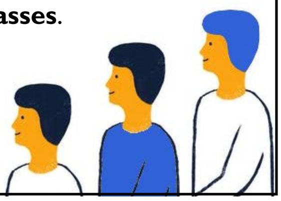
Discrete and continuous variables

- **Discrete variables**: finite number of values (e.g. "red, green, blue")
- **Continuous variables**: any value (e.g. height, time..)

A continuous variable can be **rendered discrete** by clumping it into **classes**.

E.g. short (<1,65m.)
medium (1.65m-1.80m)
tall (>1.80m)



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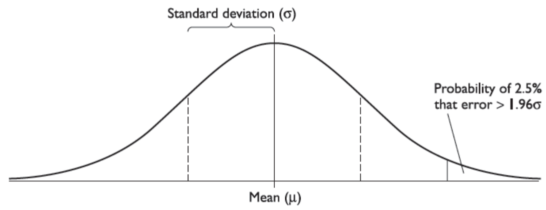
Normal Distribution and Parametric tests

If the form of the data follows a **known distribution** (**parametric tests**) then special and **more powerful statistical** tests can be used.

Most common: **normal distribution**.



Histogram of normally distributed errors

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Non-Parametric Tests and Contingency Tables

Parametric:
assume normal distribution;
robust, powerful

Non-parametric:
do not assume normal distribution;
less powerful, more reliable

Contingency table:
classify data by discrete attributes, count number of data items in each group

		sex	
		male	female
preferred interface	A	34	28
	B	26	44

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Typical Questions in Data Analysis

Is there a difference?
Question: "*Is system A better than system B?*"
Answer: "*We are 99% certain that..* "

How large is the difference? (averages)
"*Selection within system A is 260 ms faster than within system B*"

How accurate is the estimate? (standard deviation)
"*We are 95% certain that the difference in response time is between 230 and 290 ms..*"

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